Lecture 2

Solutions and Initial Value Problems

The general form for an n-th order ordinary differential equation (ODE) with independent variable x and dependent variable y can be expressed as

$$F\left(x, y, y', y'', \ldots, y^{(n)}\right) = 0 \tag{2.1}$$

ind dep

where the left hand side of the equation F represents an expression involving its arguments. We can solve for the highest-order derivative $y^{(n)}$ and rewrite this equation as

$$y^{(n)} = f(x, y, \ldots, y^{(n-1)}) \tag{2.2}$$

$\frac{dP}{dt} = kP = f(t, P)$ order 1

Definition: A function $\varphi(x)$ when substituted for $y(x)$ in (2.1) is an **explicit solution** if $\varphi(x)$ satisfies

$\varphi(x) = \sim$

$$F(x, \varphi(x), \varphi'(x), \ldots, \varphi^{(n)}(x)) = 0$$

for every x in some open interval I.

Example: Consider the equation $\frac{dy}{dx} = xy^{1/2}$. Show that $\varphi(x) = \frac{x^4}{16}$ is an explicit solution on $(-\infty, \infty)$.

$\varphi(x) = \frac{x^4}{16}$

$\underbrace{\frac{dy}{dx} - xy^{1/2}}_{F(x, y, \frac{dy}{dx})} = 0$

$F(x, y, \frac{dy}{dx}) = 0$

Is it true that $F(x, \varphi(x), \varphi'(x)) \stackrel{?}{=} 0$

Check: $\varphi'(x) = \frac{x^3}{4}$ substitute: $\varphi'(x) - x(\varphi(x))^{1/2}$

$= \frac{x^3}{4} - x\left(\frac{x^4}{16}\right)^{1/2} = \frac{x^3}{4} - x\left(\frac{x^2}{4}\right)$

$= \frac{x^3}{4} - \frac{x^3}{4} = 0$

So $\varphi(x)$ is an explicit solution.

Not all solutions to an equation are necessarily explicit solutions. In some instances we can expect a solution to be in terms of both the dependent and independent variable.

ind dep

Definition: The implicit function $G(x, y) = 0$ is an **implicit solution** to (2.1) on the interval I if it defines one or more explicit solutions on I.

$y(x)$ satisfies (2.1)

Example: Show that $\underbrace{x^2 + y^2 = 4}_{\text{circle}}$ is an implicit solution to the equation $\dfrac{dy}{dx} = -\dfrac{x}{y}$ on the interval $(-2, 2)$.

(*)

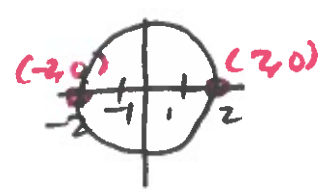
To show $x^2+y^2=4$ is an implicit solution we differentiate it implicitly with respect to the independent variable x, and rewrite it in the same form as (*)

$$\frac{d}{dx}(x^2+y^2=4) \Rightarrow 2x + 2y\frac{dy}{dx} = 0$$

$$\Rightarrow 2y\frac{dy}{dx} = -2x$$

$$\Rightarrow \frac{dy}{dx} = \frac{-2x}{2y} = -\frac{x}{y} \quad \checkmark \quad (*)$$

So $x^2+y^2=4$ is an implicit solution to (*)



Here $x^2+y^2=4$, $y(x)$ has domain $[-2,2]$ x-values

at $x=-2,2$, $y=0$ and $\dfrac{dy}{dx} = -\dfrac{x}{y}$ is undefined when $y=0$.

So this is a solution on $(-2,2)$.

We will explore how to solve (*) in Module 1 Lectures 4-7.

Note how in the last example, any circle $x^2 + y^2 = r^2$ is an implicit solution to the equation. So the family of curves $x^2 + y^2 = C$ for $C > 0$ satisfies the equation $\dfrac{dy}{dx} = -\dfrac{x}{y}$.

In general, when solving a 1st order differential equation $F(x, y, y') = 0$ we will obtain a family of solutions $G(x, y, C) = 0$ containing one arbitrary constant C. In the last example we have $G(x, y, C) = x^2 + y^2 - C = 0$.

This idea extends to n-th order differential equations $F(x, y, y', \ldots, y^{(n)}) = 0$. We can expect to obtain a family of solutions $G(x, y, C_1, \ldots, C_n) = 0$ containing n arbitrary constants $C_1, C_2, \ldots, C_n$. general solution

Example: Consider the 2nd Order equation

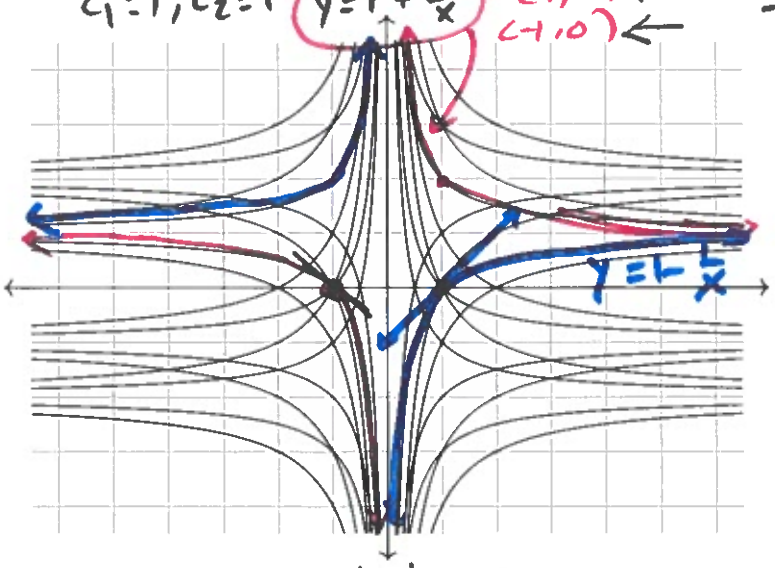
power 1

$$x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = 0 \qquad \text{linear} \tag{2.3}$$

no y's

Verify that the function $y = C_1 + \dfrac{C_2}{x}$ is a general solution to (2.3).

$C_1 = 1, C_2 = 1$ $\quad y = 1 + \frac{1}{x}$ $\quad (1,2)$ $\quad (-1,0)$



Graph of solutions for $C_1 = \pm 1, \pm 2$, $C_2 = \pm 1, \pm 2$

Check: $\dfrac{dy}{dx} = 0 - \dfrac{C_2}{x^2}$

$\dfrac{d^2y}{dx^2} = \dfrac{2C_2}{x^3}$

Substitute:

$$x\left(\frac{d^2y}{dx^2}\right) + 2\frac{dy}{dx} = \cancel{x}\left(\frac{2C_2}{x^{\cancel{3}\,2}}\right) + 2\left(-\frac{C_2}{x^2}\right)$$

$$= \frac{2C_2}{x^2} - \frac{2C_2}{x^2} = 0 \checkmark$$

So $y = C_1 + \dfrac{C_2}{x}$ is a general solution.

If we know pt and slope 2nd order then we get a specific $y(1) = 2$, $y'(-1) = -1 \Rightarrow y = 1 + \frac{1}{x}$ (particular solution)

We will explore how to solve (2.3) in Module 2.

Definition: An n-th order **Initial Value Problem** (IVP) has the following form:

$$\frac{d^n y}{dx^n} = f(x, y, y', \ldots, y^{(n-1)}) \leftarrow \text{nth order DE}$$

subject to the conditions (at x_0)

$$\text{derivatives}\left\{\begin{aligned} y(x_0) &= y_0 \quad \text{pt} \\ y'(x_0) &= y_1 \\ &\vdots \\ y^{(n-1)}(x_0) &= y_{n-1} \end{aligned}\right\} \text{initial conditions}$$

where x_0 is in the interval I and $y_0, y_1, \ldots, y_{n-1}$ are given constants. A solution to the IVP is called a **particular solution**.

Example: Find a particular solution to the IVP $x_0 = 1$

$$\underbrace{x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = 0}_{\text{Previous Example}}, \quad \underset{y_0}{y(1) = 0}, \quad \underset{y_1}{\frac{dy}{dx}(1) = 1}.$$

General solution: $y \overset{(*)}{=} c_1 + \frac{c_2}{x}$, $\frac{dy}{dx} \overset{(**)}{=} -\frac{c_2}{x^2}$

use $\begin{cases} y(1) = 0 \\ \frac{dy}{dx}(1) = 1 \end{cases}$ to find c_1 and c_2

$0 = y(1) \overset{(*)}{=} c_1 + \frac{c_2}{1} = c_1 + c_2 \Rightarrow c_1 = -c_2 = -(-1) = 1$

$1 = \frac{dy}{dx}(1) \overset{(**)}{=} -\frac{c_2}{1^2} = -c_2 \Rightarrow \boxed{c_2 = -1}$ $\boxed{c_1 = 1}$

So the particular solution is $y = 1 - \frac{1}{x}$

Example: Show that $y = \sin(x) + \cos(x)$ is a particular solution to the IVP

$$y'' + y = 0, \quad y(0) \overset{(1)}{=} 1, \quad y'(0) \overset{(2)}{=} 1$$

First, we show $y = \sin(x) + \cos(x)$ is a solution.

$$\Rightarrow y' = \cos(x) - \sin(x)$$

$$\Rightarrow y'' = -\sin(x) - \cos(x)$$

So $y'' + y = (-\sin(x) - \cos(x)) + (\sin(x) + \cos(x))$

$$= 0$$

$$y(0) = \sin(0) + \cos(0) = 0 + 1 \overset{(1)}{=} 1 \checkmark$$

$$y'(0) = \cos(0) - \sin(0) = 1 - 0 \overset{(2)}{=} 1 \checkmark$$

Yes, it is a particular solution.

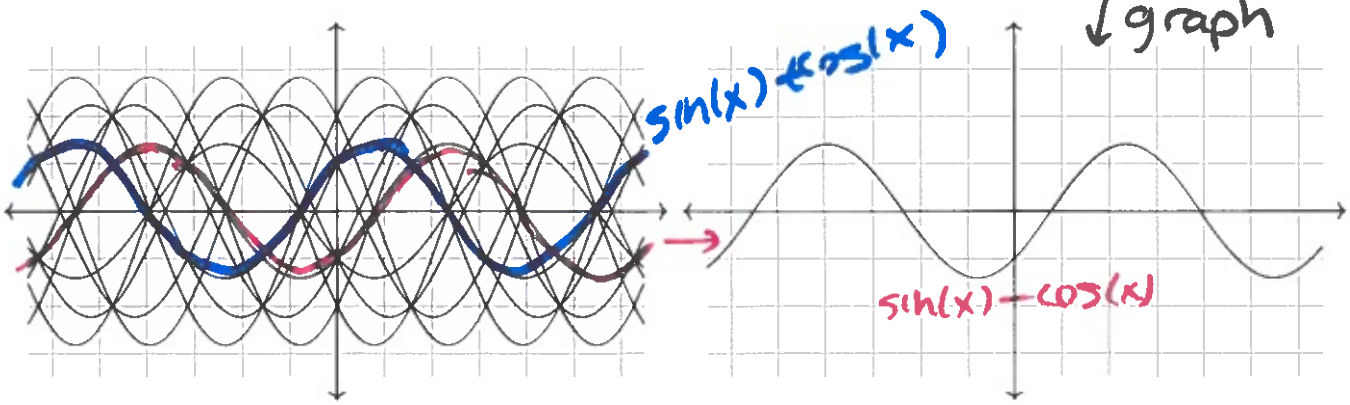
If we use a different set of initial conditions

such as $y(0) = -1$, $y'(0) = 1$

Then we get a different particular solution

$y = \sin(x) - \cos(x)$

↓graph



General solution

$$y = C_1\cos(x) + C_2\sin(x)$$

E-U

Theorem (Existence and Uniqueness of Solution): Given the IVP

– 1st order

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0 \tag{2.4}$$

condition

assume that $f(x, y)$ and $\frac{\partial f}{\partial y}$ are continuous in a rectangle

check

if (x_0, y_0) $f, \frac{\partial f}{\partial y}$ exist at (x_0, y_0)

dep

$$R = \{(x, y) | a < x < b, c < y < d\}$$

that contains (x_0, y_0). Then the IVP has a *unique* solution $\varphi(x)$ in some interval $(x_0 - \delta, x_0 + \delta)$ with $a \leq \delta \leq b$.

Example: Does the Existence and Uniqueness Theorem (E-U) guarantee a solution to the following IVPs?

1. $\frac{dx}{dt} + \cos(x) = \sin(t), \quad x(\pi) = 0$ $\rightarrow (\pi, 0)$

dep: x ind: t

$\Rightarrow \frac{dx}{dt} = \underbrace{\sin(t) - \cos(x)}_{f(t,x)}$

look at $\begin{cases} f(t,x) = \sin(t) - \cos(x) \\ \frac{\partial f}{\partial x}(t,x) = 0 + \sin(x) \end{cases}$

dep

Both $f(t,x)$ and $\frac{\partial f}{\partial x}(t,x)$ exist at $(\pi, 0)$ so they are continuous
t x

Yes, E-U guarantees a unique solution.

2. $\frac{dy}{dx} = \underbrace{3x - (y-1)^{1/3}}_{f(x,y)}, \quad y(2) = 1$ $(2, 1)$

dep: y ind: x

$$\begin{cases} f(x,y) = 3x - (y-1)^{1/3} \\ \frac{\partial f}{\partial y}(x,y) = 0 - \frac{1}{3}(y-1)^{-2/3} = \frac{-1}{3(y-1)^{2/3}} \end{cases}$$

$f(2,1) = 6$ exists, but $\frac{\partial f}{\partial y}$ does not exist at $y=1$

so not guaranteed a solution by E-U.

May have a solution, not guaranteed.